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WORD-ORDER TRANSFERENCE BETWEEN LATIN AND GREEK: THE RELATIVE POSITION OF THE ACCUSATIVE DIRECT OBJECT AND THE GOVERNING VERB IN CASSIUS DIO AND OTHER GREEK AND ROMAN PROSE AUTHORS

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THROUGH statistical analysis of a great number of sentences, we confirm one major difference between Latin and Greek word order, and we explore the ramifications of this observation for some possible cases of word-order transference between Latin and Greek. The difference to which we refer concerns the positioning of the accusative direct object before or after the verb governing it. That there is a significant difference in the Greek and Latin distributions will come as no surprise to Classical linguists: it has long been observed that Latin has a greater tendency to place the verb at the end of the clause than does Greek.[1] From this fact alone one might predict that the direct object in Latin is more likely to precede than to follow the verb on which it depends than is the case in Greek—although, logically, this need not be the case. In our study we tested this prediction empirically by tabulating the direct object distributions in sixty passages written by fifteen Latin and ten Greek prose authors. Each passage was randomly selected in the text of an author. We analyzed the first one hundred direct objects in the accusative

[1] See, e.g., Linde 1923; Schwyzer and Debrunner 1939–53: 2.695 (with older literature); Marouzeau 1953: 44 ff.; Humbert 1954: 92 ff.; Dover 1960: 25 ff.

case that were encountered, tabulating those that occurred before (DO before) and those after (DO after) the verbs that govern them.[2] The raw data can be found at the end of this article in Appendix I.

For classification of the raw data into groups we relied upon cluster analysis. This is a popular numerical method of partitioning data sets from individuals or other objects of study developed in the biological sciences for the purpose of obtaining objective and stable classifications. Whereas other methods of discrimination begin by partitioning the data into predetermined groups, the purpose of cluster analysis is to discover natural groupings (or, "clusters") of the individuals or data sets. Both approaches have their uses, but cluster analysis suits our needs because we want to test the hypothesis that Greek and Latin texts will naturally and consistently clump together into homogeneous groups if measured by the rate at which they put the DO before and after the governing verb.

Essential to cluster analysis is an ability to quantify the characteristics of individuals or other data sets that we wish to categorize. The concept of *cluster* is dependent on a measurement of the relative distances between values displayed by the various individuals or data sets under study. Values that, when graphed, coincide or are close indicate that the individuals or data sets under study can be clumped together in the same cluster.

There are different techniques for measuring distance and for

[2] In several passages, somewhat more or fewer than 100 cases were collected (the lowest number was 96; the highest, 101). The results from these passages were therefore scaled to be comparable to those for other passages. The scaled passages can be easily recognized in Appendix I because they have fractional values. As for the tabulation of direct objects, the following principles were applied:

(1) For the sake of uniformity and to simplify matters, only cases represented by nouns or pronouns (including substantives) in the accusative case were counted as direct objects. Not counted as direct objects were constructions like indirect statement or relative clauses; nouns and pronouns dependent on verbs requiring the genitive, dative, or ablative; gapped constructions in which the verb or direct object was implied but not explicitly given. Direct objects within indirect statements, relative clauses, etc. were counted.

(2) In relative clauses, relative pronouns in the accusative introducing the clause were counted as direct objects within the clauses.

(3) When there were many direct objects, only the first three were counted.

(4) In the case of apposition, both members were counted (unless, of course, they were parts of a name: e.g., *Gaium Caesarem*).

determining which objects of study belong to which clusters.[3] In this paper, we use two popular nonhierarchical and hierarchical clustering techniques that are commonly available in statistics packages today. The data sets we study are sentences in texts of Greek and Roman authors; the quantifiable feature we tabulate is the percentage of direct objects positioned before and after the governing verb.

Of the two clustering techniques, we mainly relied on *K*-means clustering, an example of nonhierarchical analysis. In *K*-means clustering, the cases are assigned to clusters in a process that calculates how well the mean values of each newly considered case correspond to the overall means (or, centroids) of all the passages in the existing clusters.[4] This form of cluster analysis was chosen because its strength is the determination of natural partitions in a data set, whereas hierarchical techniques are better at uncovering subdivisions within the partitions and the "nearest neighbors."[5] Since we wanted to test whether the data divided naturally into two groups, one composed of Latin texts, the other of Greek texts, *K* (the initial number of clusters) was set at 2.[6] As the following table shows, the passages did divide nicely into two clusters, one Greek and the other mainly Latin:

CLUSTER 1: MEMBERS

CASE	DISTANCE	CASE	DISTANCE
Varro *Res Rust.*	9.18	Ammianus Marcellinus 18	1.18
Cicero *De Off.* 3a	4.18	Pliny (Jr.) *Epist.*	9.18
Cicero *De Nat. Deor.*	1.82	Pliny the Elder *NH* 36	6.65

[3] On the basic concepts of cluster analysis used in this and the preceding paragraph, see Everitt 1993: 1–10.

[4] On *K*-means clustering, see Affifi and Clark 1984: 394–397. The algorithm is described on pages 394–395 as follows: "(1) Divide the data into *K* initial clusters. The members of these clusters may be specified by the user or may be selected by the program, according to an arbitrary procedure. (2) Calculate the means or centroids of each of the *K* clusters. (3) For a given case, calculate its distance to each centroid. If the case is closest to the centroid of its own cluster, leave it in that cluster; otherwise, reassign it to the cluster whose centroid is closest to it. (4) Repeat step 3 for each case. (5) Repeat steps 2, 3, and 4 until no cases are reassigned."

[5] Cf. Everitt 1993: 55: "in a hierarchic classification the data are not partitioned into a particular number of classes or clusters at a single step."

[6] On the logic of setting *K* at its "natural" value in an initial analysis of the data, see Affifi and Clark 1984: 400.

Cicero *Brutus*	2.18	Ammianus Marcellinus 22	1.18
Cicero *De Off.* 2a	6.18	SHA *Marcus*	0.18
Cicero *De Off.* 1b	2.18	Caesar *BG* 3	13.82
Cicero *De Off.* 1a	3.18	Tacitus *Ann.*	5.82
Cicero *De Leg.*	7.18	Caesar *BG* 1	12.82
Cicero *De Off.* 3b	1.82	Livy 21	4.83
Cicero *De Off.* 2b	0.18	Livy 3	3.41
Pliny the Elder *NH* 34	10.55	Tacitus *Hist.*	9.82
Aulus Gellius *Misc.* 1 (Greek sources)	3.82	Celsus *De Med.* 2	3.82
		Cassius Dio 36	1.18
Cicero *Ad Att.*	2.18	Cato *De Agric.*	12.82
Cicero *Cat.*	8.18	Cassius Dio 40	1.82
Seneca *Epist.*	2.18	Augustus *Res Gestae* (Latin)	6.82
Aulus Hirtius *BG* 8	1.18	Augustus *Res Gestae* (Greek)	2.82
Aulus Gellius *Misc.* 2 (Latin sources)	2.82	Tacitus *Agric.*	4.61
		Cassius Dio 59	6.51
SHA *Hadrian*	0.18	Cassius Dio 54	7.01
Seneca *De Clem.*	2.77		

CLUSTER 1: SUMMARY STATISTICS

VARIABLE	MINIMUM	MEAN	MAXIMUM	ST.DEV.
DO After	3.00	16.82	27.37	6.08
DO Before	72.63	83.18	97.00	6.08

CLUSTER 2: MEMBERS

CASE	DISTANCE	CASE	DISTANCE
Plutarch *Dinner*	2.90	Plutarch *Alcib.*	6.10
Plutarch *Alex.*	0.10	Thucydides 2	4.90
Xenophon *Anab.* 1	18.10	Marcus Aurelius 3	12.90
Polybius 2	11.10	Marcus Aurelius 8	3.90
Herodotus 2	0.10	Plutarch *Themist.*	2.90
Herodotus 1	10.10	Plutarch *Quaest. Rom.*	2.90
Polybius 1	8.10	Dionysius of Halicarnassus *Rom. Ant.* 2	5.29
Plutarch *Cicero*	3.90		
Plutarch *Sertorius*	10.90	Thucydides 7	0.24
Musonius Rufus	7.68	Plutarch *Camillus*	5.90
		Thucydides 5	2.10
		Plutarch *Quaest. Graec.*	6.90

CLUSTER 2: SUMMARY STATISTICS

VARIABLE	MINIMUM	MEAN	MAXIMUM	ST.DEV.
DO After	32.00	44.90	63.00	7.54
DO Before	37.00	55.10	68.00	7.54

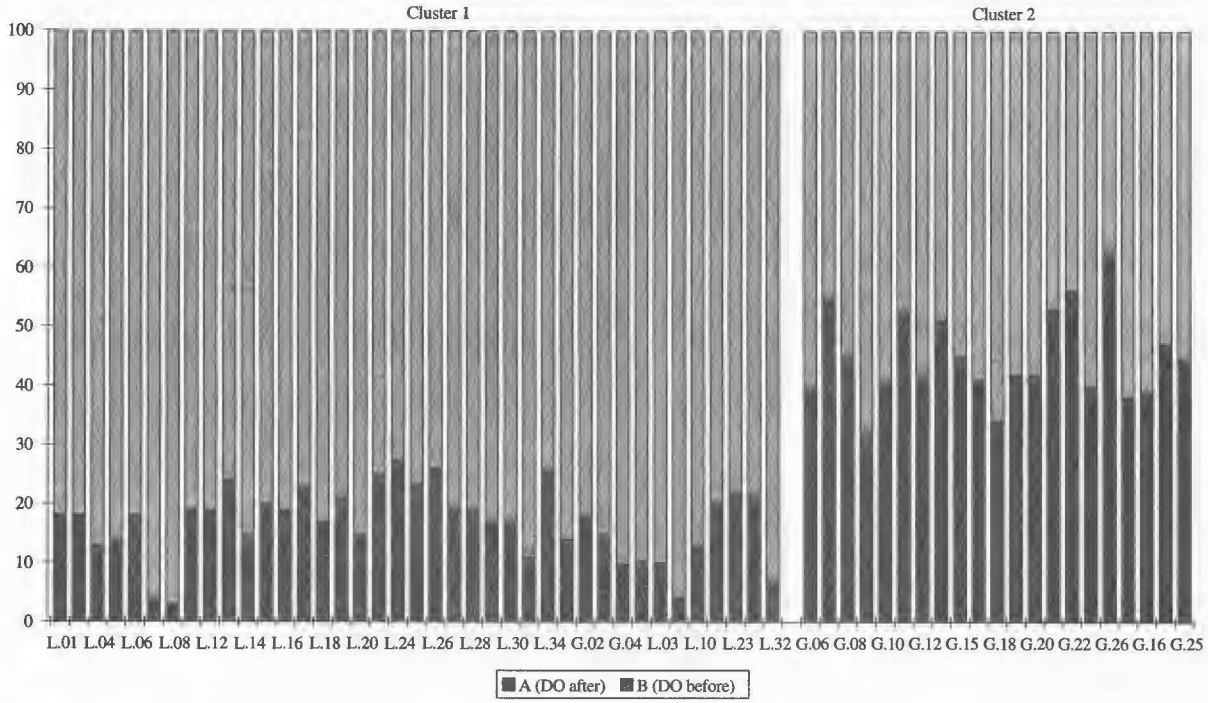
TABLE I: Cluster Analysis of Relative Positive of Direct Object (DO) Before and After Its Governing Verb (*K*-means; division into 2 groups). See Appendix I for details about passages and editions used.

The first cluster has 39 passages, all but five of which are Latin. These five cases are interesting and will be discussed below. Cluster 1 is characterized by a relatively high mean (83.18%) for the position of direct object before the governing verb, ranging from a minimum value of 72.63% to a high of 97%. The high value of this mean is what previous linguistic research led us to expect. All 21 passages in Cluster 2 are Greek. Typical of passages in this cluster is a greater balance between direct objects before (55.10%) and after (44.90%) the governing verb. For the position after the verb, the minimum value we find is 32% and the maximum value is 63%. The bar graph of Figure A shows the clearcut division of our passages into two clusters, one (the "Latin" cluster) with a high value of direct objects before the verb and a low value of direct objects after; the other (the "Greek" cluster) with a more evenly balanced distribution of direct objects. In Figure A, the percentage of occurrence of direct objects before (gray) and after (black) the governing verb in a passage can be read on the vertical axis. The horizontal axis gives the numeric code of each text in the database.[7] For example, on the extreme left, we have the values for Ammianus Marcellinus Book 18 (L.1; DO after = 18%; DO before = 82%). On the extreme right, we have those for Thucydides Book 7 (G.25; DO after = 45%; DO before = 55%). We can see that there is a natural division into two clusters: passages on the righthand side, in which the direct object is almost as likely to occur before as after the verb; and those to the left, in which the direct object is much more likely to occur before the verb. The passages on the right (Cluster 2) are all from Greek texts; those on the left (Cluster 1) are primarily from Latin.

We do not have to rely on our visual impression of the graph to grasp that the division into two clusters is statistically significant. The

[7] We use numerical reference to save space. For a concordance of the numeric codes and the texts to which they refer, see Appendix I.

FIGURE A: Bar Graph of Data on Table I. Horizontal axis is the code for author and passage. Vertical axis is percentage of accusatives after and before governing verb. For the numeric codes identifying the texts, see Appendix I.



summary statistics for the *K*-means test show that there is an extremely low probability ($p<0.001$) that the sharp division of passages into two clusters occurs by chance. The proximity of the means of each passage to the centroids of its cluster can be gauged from the value given on the table under Distance. Thus, in our first cluster, the values of 0.18 for the passages from the *Lives* of Hadrian (DO before = 83%) and Marcus Aurelius (DO before = 83%) in the *Historia Augusta* and from Cicero's *De Officiis* 2b (2.36–55; DO before = 83%) indicate that the mean values of direct object before and after the governing verb found in these passages most closely approximates the centroids of the cluster as a whole (83.18%). The passage least characteristic of the cluster is Caesar's *Bellum Gallicum* 3 (DO before = 97%; Distance = 13.82).

A recent article by Pinkster suggested to us that our results might differ if we distinguished between direct objects of the main verb in *main clauses* (MC) and those that are objects in *other constructions* (OC) such as subordinate clauses, participial and infinitive phrases, etc.[8] A detailed investigation of constructions other than direct objects of main verbs would require more data than we collected, but we do have sufficient information at least to gauge whether the consistent breakdown into Latin and Greek groups is maintained if the cases are categorized by MC and OC. We once again used *K*-means clustering, increasing the number of clusters (*K*) until we reached a point at which a cluster contained only one passage. This we considered too fine a degree of analysis. That happened when $K = 5$, suggesting that our data naturally breaks down into four clusters:

SUMMARY STATISTICS

VARIABLE	BETWEEN SS	DF	WITHIN SS	DF	F-RATIO	PROBABILITY
MCB	8658.233		3809.04	59	44.70	< 0.0005
MCA	2813.183		1229.14	59	45.01	< 0.0005
OA	4422.423		1749.99	59	49.70	< 0.0005
OB	7924.673		3697.08	59	42.16	< 0.0005

[8] Pinkster 1991: 69.

CLUSTER I: MEMBERS

CASE	DISTANCE	CASE	DISTANCE
Cicero *De Off.* 3a	6.99	Cicero *Cat.*	5.94
Cicero *De Nat. Deor.*	10.14	Seneca *Epist.*	1.72
Cicero *Brutus*	7.59	Aulus Hirtius *BG* 8	3.00
Cicero *De Off.* 2a	5.93	Aulus Gellius *Misc.* 2	
Cicero *De Off.* 1b	6.35	(Latin sources)	2.55
Cicero *De Off.* 1a	4.70	SHA *Hadrian*	4.68
Cicero *De Leg.*	4.60	Seneca *De Clem.*	4.24
Cicero *De Off.* 3b	4.32	Ammianus Marcellinus 18	4.51
Cicero *De Off.* 2b	3.38	Pliny (Jr.) *Epist.*	8.99
Pliny the Elder *NH* 34	5.63	Pliny the Elder *NH* 36	7.16
Aulus Gellius *Misc.* 1		Ammianus Marcellinus 22	5.01
(Greek sources)	2.86	SHA *Marcus*	5.49
Cicero *Ad Att.*	1.13	Caesar *BG* 3	8.20
		Tacitus *Ann.*	7.61
		Caesar *BG* 1	8.59

CLUSTER I: SUMMARY STATISTICS

VARIABLE	MINIMUM	MEAN	MAXIMUM	ST.DEV.
MCB	19.00	34.06	48.00	7.83
MCA	3.00	10.25	20.00	4.70
OA	0.00	7.96	19.00	4.25
OB	34.00	47.74	62.00	6.86

CLUSTER II: MEMBERS

CASE	DISTANCE	CASE	DISTANCE
Livy 21	7.07	Cassius Dio 40	3.61
Livy 3	6.31	Augustus *Res Gestae* (Latin)	2.27
Tacitus *Hist.*	8.10	Augustus *Res Gestae* (Greek)	4.01
Celsus *De Med.* 2	5.02	Tacitus *Agric.*	8.03
Cassius Dio 36	3.47	Cassius Dio 59	8.98
Cato *De Agric.*	6.98	Cassius Dio 54	9.78

CLUSTER II: SUMMARY STATISTICS

VARIABLE	MINIMUM	MEAN	MAXIMUM	ST.DEV.
MCB	48.45	60.08	75.49	8.48
MCA	3.00	9.37	16.49	4.08
OA	0.00	4.33	12.24	3.55
OB	13.27	26.22	40.00	8.42

CLUSTER III: MEMBERS

CASE	DISTANCE	CASE	DISTANCE
Plutarch *Dinner*	8.55	Plutarch *Sertorius*	7.92
Plutarch *Alex.*	6.58	Musonius Rufus	4.42
Xenophon *Anab.* 1	8.87	Plutarch *Alcib.*	4.18
Polybius 2	10.80	Thucydides 2	7.49
Herodotus 2	4.07	Marcus Aurelius 3	7.91
Herodotus 1	5.32	Marcus Aurelius 8	5.18
Polybius 1	4.65	Plutarch *Themist.*	5.12
Plutarch *Cicero*	4.06	Plutarch *Quaest. Rom.*	7.41
		Dionysius of Halicarnassus *Rom. Ant.* 2	8.38

CLUSTER III: SUMMARY STATISTICS

VARIABLE	MINIMUM	MEAN	MAXIMUM	ST.DEV.
MCB	14.00	23.61	37.62	6.02
MCA	5.00	17.03	25.00	6.16
OA	16.00	28.51	47.00	7.36
OB	20.00	30.85	44.00	7.61

CLUSTER IV: MEMBERS

CASE	DISTANCE
Thucydides 7	3.51
Plutarch *Camillus*	3.81
Thucydides 5	2.48
Plutarch *Quaest. Graec.*	4.74

CLUSTER IV: SUMMARY STATISTICS

VARIABLE	MINIMUM	MEAN	MAXIMUM	ST.DEV.
MCB	37.86	41.97	50.00	4.72
MCA	28.00	32.97	37.86	4.08
OA	6.80	9.20	11.00	1.85
OB	12.00	15.87	21.00	3.61

TABLE II: Cluster Analysis of Relative Positive of Direct Object Before (B) and After (A) Its Governing Verb in Main Clauses (MC) and Other Constructions (O) (*K*-means; division into 4 groups)

In Table II, we show our tabulations for the position of direct objects. MCB is our abbreviation for a direct object occurring before the verb in a main clause. OB indicates any other kind of syntactic construction in which the direct object occurs before the verb governing it. This might be a subordinate clause, a participial construction, an infinitive clause, etc. MCA stands for a direct object occurring after the verb in a main clause. OA indicates any other kind of syntactic construction in which the direct object occurs after the verb governing it.

From the table, we can see that when subjected to a *K*-means cluster analysis, our 60 passages can be assigned to four groups—two mainly Latin and two exclusively Greek. Clusters I and II are almost entirely composed of Latin authors and are distinguished from the two Greek groups (Clusters III and IV) by the fact that they have a much higher percentage of direct objects that come before their verb (i.e., MCB and OB). Contrariwise, the Greek Clusters III and IV have a much higher percentage of direct objects that follow their verb (i.e., MCA and OA). As the summary statistics at the head of Table II show, there is very low probability that the assignment of the texts to four clusters reflects the operation of chance alone. The variables MCB, MCA, OB, and OA are very useful in separating the clusters.

Table III consolidates the summary statistics for the individual clusters from Table II, showing that our Cluster I differs from Cluster II analogously to the way in which Cluster III is different from Cluster IV. In Cluster I, the mean value for DO in the OB position is greater than

that in the MCB position; Cluster II has the reverse.[9] Similarly, Cluster III has more direct objects in the OA than the MCA position; and Cluster IV has the reverse. Figure B permits us easily to visualize these relationships.

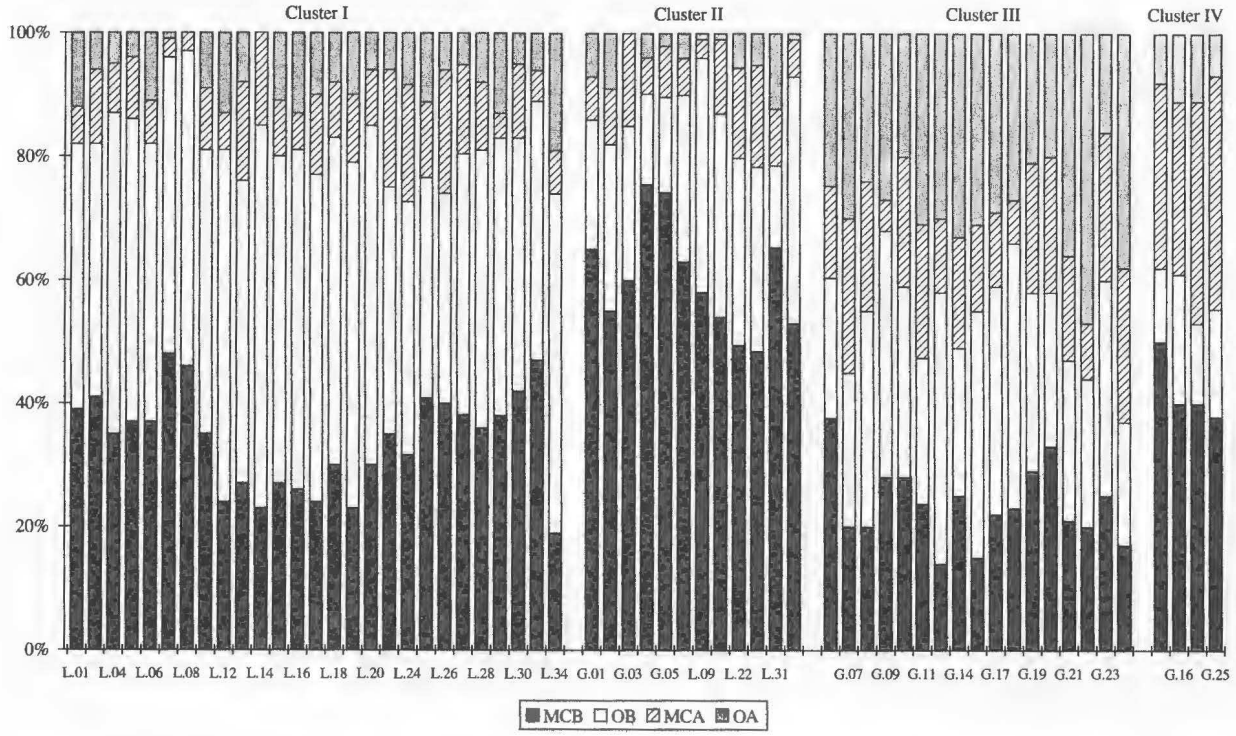
VARIABLE	MINIMUM	MEAN	MAXIMUM	ST.DEV.
GROUP I (Latin): More OB than MCB				
MCB	19.00	34.06	48.00	7.83
MCA	3.00	10.25	20.00	4.70
OA	0.00	7.96	19.00	4.25
OB	34.00	47.74	62.00	6.86
GROUP II (Latin): More MCB than OB				
MCB	48.45	60.08	75.49	8.48
MCA	3.00	9.37	16.49	4.08
OA	0.00	4.33	12.24	3.55
OB	13.27	26.22	40.00	8.42
GROUP III (Greek): More OA than MCA				
MCB	14.00	23.61	37.62	6.02
MCA	5.00	17.03	25.00	6.16
OA	16.00	28.51	47.00	7.36
OB	20.00	30.85	44.00	7.61
GROUP IV (Greek): More MCA than OA				
MCB	37.86	41.97	50.00	4.72
MCA	28.00	32.97	37.86	4.08
OA	6.80	9.20	11.00	1.85
OB	12.00	15.87	21.00	3.61

TABLE III: Summary Statistics for Groups I–IV

Of our twenty-four authors, sixteen are represented by two or more texts. Examination of the cluster analysis shows that most of these sixteen authors' passages consistently appear in just one cluster. Only one Latin author (Tacitus; L.31–L.34) and two Greek authors (Plutarch;

[9] There are two exceptions (Pliny *NH* 36 and Pliny the Younger *Epist.*), which are classified into Cluster I despite having more MCB than OB. This occurs because their mean OB falls more than 2 standard deviations below the mean value for OB of Cluster II, and their mean MCB is close to the mean MCB for Cluster I.

FIGURE B: Bar Graph of Data in Table II. Horizontal axis is the code for author and passage. Vertical axis is percentage of accusatives before (MCB) and after (MCA) verb in main clause, and before (OB) and after (OA) verb in other constructions.



G.12–G.20 and Thucydides; G.23–G.25) have works appearing in two clusters. Within each cluster, we can determine how closely related the texts are to each other by hierarchical cluster analysis. The first step is to divide the texts once again into clusters, though this time a different technique of partition is used. Once again, four clusters (two Latin, and two Greek) emerge, as follows:

(A) Thucydides (5, 7) and Plutarch (*Camillus*, *Quaest. Gr.*)
(B) Augustus, Cassius Dio (36, 54, 59), and Tacitus (*Agricola*)
(C) Marcus Aurelius, Dionysius of Halicarnassus, Herodotus, Musonius, Plutarch (the remaining works), Polybius, Thucydides (2), and Xenophon
(D) Ammianus Marcellinus, Caesar, Cassius Dio (40), Cato, Celsus, Cicero, Gellius, Hirtius, Pliny the Elder, Pliny the Younger, Scriptores Historiae Augustae, Seneca, Tacitus, Varro

As can be seen when the results are graphed in the form of a dendrogram (Figure C), not infrequently works by the same author are nearest neighbors in a cluster. The author with the most texts is Cicero, who is represented by eleven works in the genres of epistle, oratory, rhetoric, and philosophy. All of these works are grouped together in Cluster (D) and, as can be seen in Figure C, many are nearest neighbors. These results suggest that preferences in positioning the direct object with respect to construction type may turn out to be a useful instrument in the tool kit of authorship studies, especially if the authors under consideration for our target text for attribution belong to different clusters in a dendrogram.

Returning to our nonhierarchical form of cluster analysis, we have characterized Clusters I and II as "Latin," and Clusters III and IV as "Greek." As noted, there are two complications. As Table II shows, Cluster II has twelve passages taken from the works of seven authors. Seven passages were certainly written by native speakers of Latin (the works by Augustus [L.3], Cato [L.9], Celsus [L.10], Livy [L.22, L.23], and Tacitus [L.31–L.33]); the remaining five were written in Greek by two authors—one by the unknown translator of Augustus' *Res Gestae* (G.1) and four by Cassius Dio (G.2–G.5).[10] We note that the same five

[10] Four passages—an unusually high number for our study—were taken from different books by Dio in order to be sure that our unexpected results were generally valid for this author.

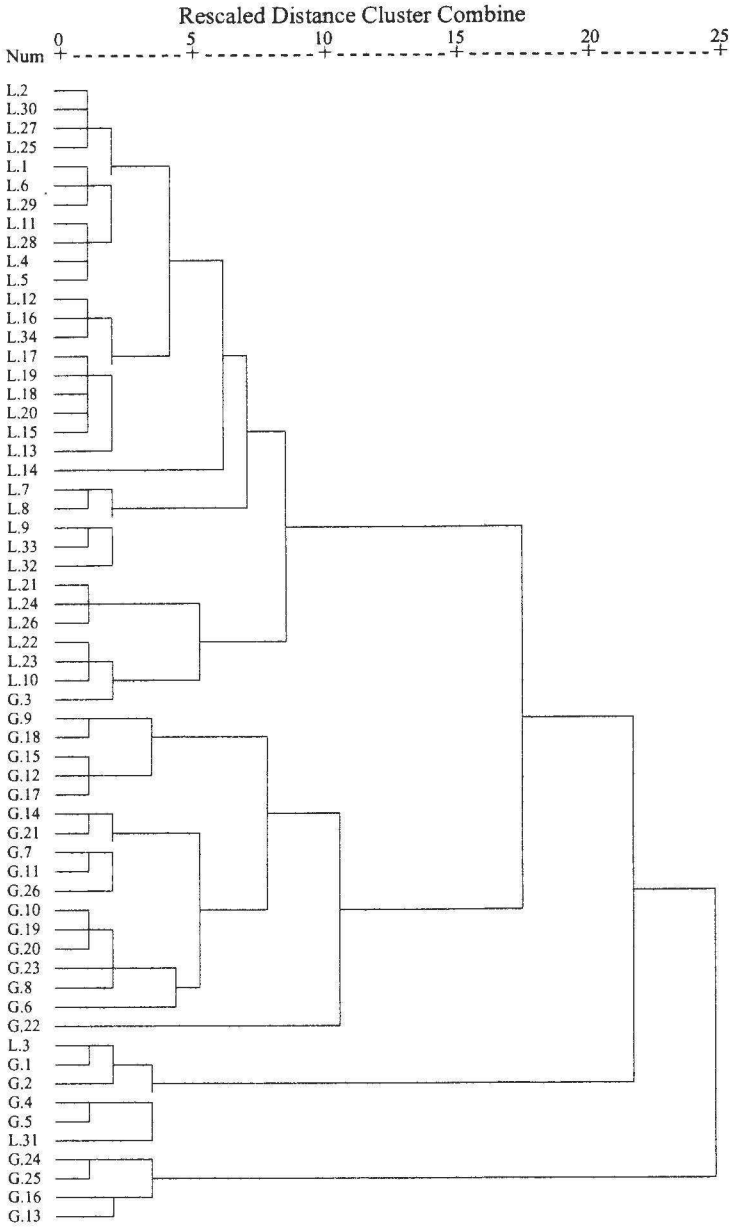


FIGURE C: Dendogram of Data in Table II.

Greek passages were classified into our first ("Latin") cluster on Table I and into our hierarchical Cluster (B) in Figure C. On Table II, the Greek texts fall into the middle of Cluster II. The average sum of MCB and OB is 87%. Cassius Dio's Greek texts range from a low of 82% to a high of 92%. The Greek translation of Augustus' *Res Gestae* weighs in at 86%. The gap between the Greek texts of Cluster II and Cluster IV (the corresponding group in which MCB exceeds OB) is considerable: the average sum of MCB and OB in Cluster IV is just 58%, with the highest values registered for Plutarch, *Quaestiones Graecae* (62%) and *Life* of Camillus (61%).

To understand what factors are most responsible for the presence of our five Greek texts in the "Latin" cluster, we can use a technique such as Principal Components Analysis (PCA). In PCA, we try to reduce the number of dimensions (i.e., variables) in a data set with only a small loss of information. Since in our case the data are 3-dimensional—as they are percentages, the fourth variable is just 100% minus the other three—there will be just three principal components, and the third one should play an insignificant role. The point of PCA is to enable us to "see" the relevant pattern of authors and works by plotting the principal component scores in the space of just the first two components, thereby obtaining a 2–dimensional picture of the data set.[11] PCA is an exploratory statistical technique in that the reduction of the descriptive complexity of a data set allows the key variables and their interrelationships to be better understood.

Figure D shows the 60 works plotted in the space of the first two principal components, using the symbol "L" for Latin works and "G"

[11] Affifi and Clark 1984: 309–310, describe PCA as follows: "principal components analysis is performed in order to simplify the description of a set of interrelated variables. In principal components analysis the variables are treated equally; i.e., they are not divided into dependent and independent variables, as in regression analysis. The technique can be summarized as a method of transforming the original variables into new, uncorrelated variables. The new variables are called the *principal components*. Each principal component is a linear combination of the original variables. One measure of the amount of information conveyed by each principal component is its variance. For this reason the principal components are arranged in order of decreasing variance. Thus, the most informative principal component is the first, the least informative is the last . . . An investigator may wish to reduce the dimensionality of the problem, i.e., reduce the number of variables without losing much of the information. This objective can be achieved by choosing to analyze only the first few principal components. The principal components not analyzed convey only a small amount of information since their variances are small."

for Greek works. The proportion of the variation in our original data set which is accounted for by the principal components is:

PC1	PC2	PC3
45.3%	38.9%	15.8%

Hence the 2-dimensional representation accounts for fully 84.2% of the original variation, a clearly acceptable simplification of the data matrix with minimal loss of information.

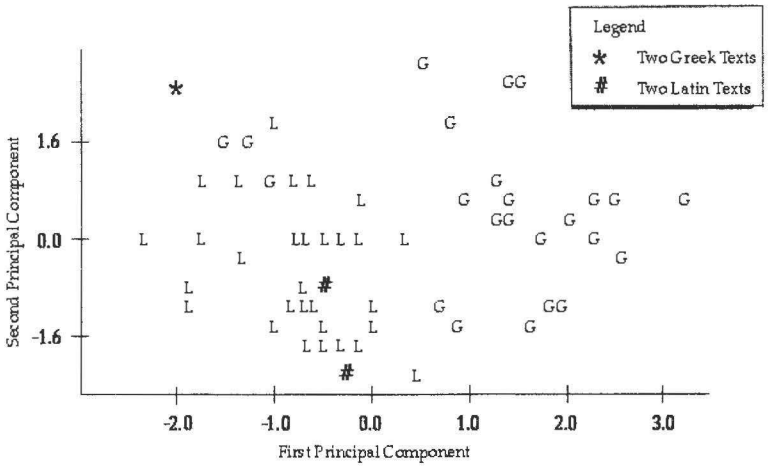


FIGURE D: Plot of Data in Table II in the space of the first two principal components. "L" = Latin texts; "G" = Greek texts.

This remarkable plot highlights the separation of the works into Latin and Greek clusters along the axis of the first principal component, which, in the present case, contrasts the MCB value against the sum of MCA + OA, with the Latin works having strong MCB measures. Apart

from five Greek works appearing in the "L" cluster, the works written in each language cluster closely together. The five anomalous works are the same Greek texts that have emerged as exceptional in our previous analyses: the four books of histories by Cassius Dio and the Greek translation of Augustus' *Res Gestae*. The PCA enables us to grasp the complex pattern in our data set and to see that the five Greek texts are unusual in that their MCB measures are very high in comparison to the sum of MCA and OA.

The presence of Greek texts in Cluster II, and the absence of Latin texts from Clusters III and IV are striking results that warrant discussion. As we have seen, PCA is a statistical technique whose power is primarily descriptive: through simplification, it enables us to see factors and patterns that would otherwise be hard to discern; however it does not offer an explanation for what it brings into view. For that, further literary, linguistic, and historical analysis is required. A solid point of departure is offered by the Greek version of the *Res Gestae*, since we know that this is a translation of the (preserved) Latin original[12]—the Greek version's nearest neighbor in our hierarchical cluster analysis (see L.3 and G.1 in Figure C). Close comparison of the two texts shows that the Greek gives practically a word-for-word rendering of the Latin version, so much so, in fact, that we cannot be certain that the translator was a native speaker of Greek, Latin, or some other language.[13] A typical paragraph (20) is printed below, facilitating a comparison of the Greek and Latin versions:

> Capitolium et Pompeium theatrum utrumque opus impensa grandi refeci
>
> Καπιτώλιον καὶ τὸ Πομπηΐου θέατρον ἑκάτερον τὸ ἔργον ἀναλώμασιν μεγίστοις ἐπεσκεύασα

[12] The text written in large letters over the first seventeen columns of the Greek version of the Monumentum Ancyranum states that the text is a translation of an original inscribed on two bronze pillars in Rome (cf. Suet. *Vita Aug.* 101).

[13] In the literature on the inscription, debate about the native language of the translator has gone on for over a century without reaching a conclusion; see Gagé 1977: 11–13 (native language is uncertain); Meuwese 1926 (translator could have been either a Roman or a Greek); T. Mommsen and G. Kaibel in Mommsen 1883: 197–202 (the translator was a Roman); Nissen 1886: 494 (the translator's native language was neither Latin nor Greek but a language of Asia); Pugliese Carratelli 1947: 79 (agrees with Nissen that translator was from Asia); Regard 1924 (native language is κοινή).

sine ulla inscriptione nominis mei. Rivos aquarum compluribus locis

ἄνευ ἐπιγραγῆς τοῦ ἐμοῦ ὀνόματος. Ἀγωγοὺς ὑδάτων ἐν πλείστοις τόποις

vetustate labentes refeci, et aquam, quae Marcia appellatur, duplicavi fonte

τῇ παλαιότητι ὀλισθάνοντας ἐπεσκεύασα καὶ ὕδωρ τὸ καλούμενον Μάρτιον ἐδίπλωσα πηγὴν

novo in rivum eius inmisso.

νέαν εἰς τὸ ῥεῖθρον αὐτοῦ ἐποχετεύσας.

In this passage, every direct object precedes the verb in both the Latin and the Greek; indeed, the Greek word order almost exactly mirrors that of the Latin original, a process that when taken to extremes is known as relexification.[14] The same technique of translation (with perhaps even less freedom) was apparently used in the Greek version of Octavian's edict concerning Seleucus of Rhosos (42–30 B.C.)[15] and Augustus' funeral oration for Agrippa. Of these two texts only the Greek translations survive, but their word order is so unnatural that scholars have not hesitated to detect the influence of the Latin original on the Greek and even (in the case of the oration for Agrippa) to attempt a back-translation.[16]

Word-for-word translation from Latin gives Greek a *color Latinus*,[17] a Latin flavor. As a method of official translation into Greek, word-for-word translation is not limited to the Augustan age but may be attested at least as late as the third century A.D. Examples include the Pisidian inscription on requisitioned transport from the beginning of the reign of Tiberius;[18] the Monumentum Ephesenum of A.D. 62;[19] and, from the

[14] See H. Zobl 1980; Andersen 1990: 62. We do not mean to imply that the Greek text always reflects the Latin original with complete fidelity. A typology of errors found with some examples is given, e.g., in Gagé 1977: 10 n. 1.

[15] Roussel 1934: 33 ff.

[16] See Haslam 1980: 193–199, esp. 195.

[17] Stroux 1928: 22; De Visscher 1940: 36.

[18] See Mitchell 1976: 106–131, esp. 106 for the date and 110 for the style of the Greek translation. Note that the word order of the Pisidian inscription follows the Latin text with greater freedom than, e.g., the *Res Gestae* translation, with occasional words and phrases relocated in the sentence.

[19] Ed. pr. in *Epigr. Anatol* 14 (1989); cf. *AE* 1989 nr. 681; *SEG* 39 (1989) nr. 1180;

years 212 and 215, the first and third edicts of Caracalla preserved in Papyrus Giessen 15.[20] Thus, this form of translation may have been a perennial feature of the Imperial chancellery. Its absence, though infrequent, is attested and may be taken, in conjunction with other linguistic evidence, as further support of theories that the documents in question were originally composed in Greek, not Latin. A case in point are the Augustan inscriptions from the agora of Cyrene. Stroux speculated on linguistic grounds other than the relative position of direct object and its governing verb that the first four documents were written in Greek, while the fifth (a *senatus consultum*) was composed in Latin. Adding the longest of Stroux's Greek documents (I) to the cluster analysis of our Table I, we find that it is classified into the Greek Cluster 2. Likewise, Stroux's Latin document (V) is assigned squarely to the Latin Cluster 1.[21]

One reason for the word-for-word style was perhaps a desire to ensure that translations of official documents were as close to the original as possible so as to avoid mistakes and misunderstandings.[22] A second factor is what might be called Roman linguistic imperialism in dealings with the Greek world. Latin remained the official language for legal and government business, even in the Greek East, until well into the third century A.D.[23] Laws and *senatus consulta* were of course composed in Latin, and in bilingual inscriptions the Latin copy generally precedes the Greek translation.[24] The use of Latin word order in the

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Lewis 1995: 248. The *editio princeps* (p. 6) speaks of how the Greek text is a "word for word" translation from the Latin.

[20] See Meyer 1910–12: 25–45 (and esp. 29, where Meyer notes the Latin word order of the Greek translation of the edict). Stein 1915: 154–156, thinks that the second edict is a translation from the Latin, too. On the styles found in Greek translations of official Roman documents see Viereck 1888; Stroux 1928: 18–43.

[21] The figures are as follows: Text I (DO after = 55%; DO before = 45%); Text V (DO after = 14%; DO before = 86%).

[22] Cf. Stroux 1928: 19; cf. Sherk 1969: 13: "*senatus consulta* were important documents, and their translation could not be left in the hands of amateurs or Greek provincials, who might deliberately or unintentionally distort the true meaning. One official source was responsible for them, and, in the light of the *color latinus* which they display, that source could only have been in Rome."

[23] See Stein 1915: 138–160; Sherk 1969: 13 writes: "the texts span a period of two hundred years, yet one sometimes feels that a single individual has done them all."

[24] Cf. Stein 1915: 152 and the literature cited by Laffi 1967: 39 n. 45. This is also true of the Pisidian inscription, published in Mitchell 1976. Stroux thought that the four Augustan edicts from Cyrene were originally written in Greek (Stroux 1928: 24–25). The

typical Greek translation of an official document may perhaps reflect yet another manifestation of the same striving for linguistic hegemony. As one scholar has put it while describing linguistic conflict in the Roman empire: Roman official texts in Greek "were literal and wooden . . . Evidently official Rome would not concede too much. Their Greek had to have a strong Latin flavor."[25]

What "official Rome" did collectively in commissioning Greek translations of official documents reflected the behavior of individual Romans in the government. As early as the second century B.C., the Roman senatorial class, for all its philhellenism and sense of linguistic inferiority vis-à-vis Greek, was uncomfortable about the use of the Greek language for public business.[26] Relevant for understanding the style of the *Res Gestae* and similar official translations are anecdotes like that about Roman officials like L. Aemilius Paullus, who, though (like Cassius Dio) fluent in Greek, addressed the Greeks in Latin at Amphipolis at the end of the Third Macedonian War.[27] Later, Cicero was criticized by his fellow senator, L. Metellus, for addressing the Syracusan senate in Greek.[28] According to Cicero (perhaps writing tongue-in-cheek), L. Lucullus wrote a Greek history containing some intentional Latinisms, "to make his readers more willing to believe that it was written by a Roman."[29]

We do not have any of Lucullus' history, so it is impossible to know just what Latinisms he used to season his style with a Roman flavor.[30] That Greeks perceived an SOV word order as characteristically Latinate is suggested by an amusing anecdote in Lucian's *Demonax*. "A man named Polybius," wrote Lucian, "quite uneducated and ungrammatical,

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situation may sometimes have been more complicated; cf. Laffi 1967: 38–39 (a calendar from the province of Asia which may represent a Latin draft by the proconsul or other official written up for publication in Greek).

[25] Bonner 1929–30: 589.

[26] Gruen 1984: 1.264–270; on the Roman sense of linguistic inferiority see Rosén 1996: 533.

[27] Gruen 1984: 1.267.

[28] Cf. Crawford 1984: 51–52; Cicero also addressed the citizens of Henna in Greek: see Crawford, 53–54.

[29] Cf. Cic. *Ad Att.* 19.10 (Shackleton Bailey): *commentarium consulatus mei Graece compositum misi ad te, in quo si quid erit quod homini Attico minus Graecum eruditumque videatur, non dicam quod tibi, ut opinor, Panhormi Lucullus de suis historiis dixerat, se, quo facilius illas probaret Romani hominis esse, idcirco barbara quaedam et soloeca dispersisse* . . . .

[30] For modern secondary literature on Latinisms in Greek, see Domingo 1979: 30.

said: 'The emperor has honored me with Roman citizenship.' 'Oh, why didn't he make you a Greek instead of a Roman?' said Demonax."[31] Polybius' word order is SOV, with the verb placed at the end of the sentence—"Romanized Greek," as one scholar has recently noted.

The stories about Polybius and Lucullus may help us to understand not only the chancellery style of the *Res Gestae* translation but the prose style of Cassius Dio as well.[32] We see no evidence that would justify explaining Dio's tendency to end clauses with a verb on diachronic grounds of Greek linguistic evolution or as a feature of the κοινή. Rather, we believe that one or both of the following two factors may explain his unusual placement of direct objects relative to the governing verb: (1) like Lucullus, Dio may have intentionally added a Latinism to his style; (2) despite his undeniable Greek ethnic background, Latin may have been Dio's preferred language for many years, especially in the period in which he composed his history. The first reason is more conscious: Dio would presumably be embellishing his text with a Latinism in order to reinforce the *Romanità* of his account of Roman history. The second is more unconscious: living in Italy for so long, Dio has fallen into a Latinate pattern of word order.

To begin with the second possibility, Dio was ethnically a Greek who identified his fatherland as Nicaea but also a second- or possibly even third-generation Roman senator by status, who rose to be praetor in 194, suffect consul in 205 or 206, and who was a member of the *consilium principis* during the reign of Septimius Severus.[33] For many years, then, Dio was immersed in life in Rome, participating in the Senate and, as a writer of a Roman history, constantly studying texts and documents written in Latin. His father was a senator, which probably meant that Dio (who was born c. 164) spent at least a portion of his

[31] Lucian *Demonax* 40: Πολυβίου δέ τινος, κομιδῇ ἀπαιδεύτου ἀνθρώπου καὶ σολοίκου, εἰπόντος, ὁ βασιλεύς με τῇ Ῥωμαίων πολιτείᾳ τετίμηκεν. Εἶθε σε, ἔφη, Ἕλληνα μᾶλλον ἢ Ῥωμαῖον πεποιήκει. Jones 1986: 96 n. 43 notes that the word order of Polybius' boast "seems to be Latin."

[32] Latinisms in Dio's prose have long been remarked, but seldom studied in any depth. See Millar 1964: 41–42; and Baldwin 1986: 479–486, esp. 481 with n. 16.

[33] On Dio's family background, and its strong ties to Rome going back to the first century A.D., see Millar 1964: 8; and on his career, 15–17. On Nicaea as Dio's fatherland, see 76[75].15.3, 80[80].5.2. On Dio's relationship with Bithynia see W. Ameling 1984, who also notes that when in Italy Dio lived in Campania, as did many other provincial Greeks working in the Roman administration, thus making for a large, Greek-speaking population in that area.

childhood in Italy. He tells us (72.4.2) that beginning with his treatment of the reign of Commodus (180–192) he was an eyewitness of the events he recounts, implying that he was living in Rome. He was to remain in Italy until 218, returning briefly to serve as consul in 229, which means that he was there during much of the period in which he wrote the *Roman History*.[34]

While Dio's habit of using a more Latin than Greek word order might even suggest that his mother tongue was Latin, this, as we will see, would be a conclusion that does not necessarily follow from our evidence. To be sure applied linguists have observed that learners of second languages often transfer features of their first language (L1) to the language they are studying (L2).[35] These features may range from the phonological to the morphological or syntactic, including word order. An example is reported by Selinker (1969). He studied the English of 132 Hebrew-speaking children in Israel and that of 31 native English speakers in the United States. As the following example shows (concerning the relative order of words for time and place in Hebrew, English, and the English interlanguage of the native Hebrew speakers), when in L2 and L1 two word orders are possible, but a different one is preferred in each language, the interlanguage reflects the preference of L1. In Selinker's example, the issue is the relative order of words for time and place in Hebrew and English. Speakers of Hebrew prefer to put time before place; English speakers do the opposite. When native Hebrew speakers learned English, they retained their natural tendency to put words for time before words for space.

Word combination	*Arrangement*	*Hebrew*	*Interlang.*	*English*
Time + place	Pl-Ti	34	24	*33
	Ti-Pl	*44	*46	4

TABLE IV. Relative order of words for time and place in utterances of native speakers of Hebrew (*Hebrew*), of English (*English*), and of the English interlanguage of native Hebrew speakers (*Interlanguage*; * = preferred word order; Pl = place; Ti = time)[36]

[34] See Barnes 1984 and Rich 1990: 1–4.

[35] The literature on transference is substantial. For a bibliography see Singh and Martohardjono 1991; for a survey of recent trends in transference theory see Odlin 1992: 176–178.

[36] Data are taken from Selinker 1969, Table 1.

Other examples are reported by Odlin involving word-order shifts from L1 to L2 in second language acquisition of Hawaiian Pidgin English, Bamboo English, Andean Spanish, and Pidgin Fijiian.[37]

But we need not hypothesize that Dio was a native Latin speaker and a mere learner of Greek (he obviously was not) to explain our data by the theory of linguistic transference. It has been noted that even highly proficient speakers of L2 may persist in borrowing features from L1 that are not present (or, in our case, we should say that are not as frequent) in L2 if they perceive that the distance between L1 and L2 is not great.[38] It is not necessary to invoke the ancient theory that Latin was a Greek dialect[39] to make plausible a claim that a writer like Cassius Dio would have viewed Latin and Greek as closely related languages. The fact is that concerning the location of the accusative direct object there was no correct and incorrect position in either language, just a distributional trend.

This is not to say that the first reason of intentional Latinism was not (also) operative. Whether or not Latin was Dio's native or preferred language, Millar observed that Dio's "identification with the Roman world is complete and unquestioned."[40] Aalders noted that Dio speaks of the Romans in the first person—a rarity among Greek authors. He calls Italy, "this land which we inhabit" (1 F 1.3), and calls the toga, the dress "we use in the Forum" (9 F 39.7). While not eschewing his Greek heritage, by virtue of his education and training, and most importantly his career as a Roman senator and historian, he was deeply immersed in the Latin literary tradition.[41] One possible explanation of Dio's Latinate placement of the direct object relative to the verb, then, is that it is very much in keeping with his self-definition as a Roman. Although far from the quasi-relexification of the official texts we have mentioned, Dio's style would have had the added advantage of giving reinforcement to his goal of Thucydidean objectivity: to his Greek reader (and it was primarily for such a reader that Dio wrote),[42] Dio may have given the

[37] Odlin 1990: 98–104. That the cases studied here should be considered examples of Second Language Acquisition and not Foreign Language Learning is a point Odlin makes at p. 114.

[38] Corder 1983: 95; Kellerman 1983: 114.

[39] See Giomini 1953; Gabba 1963.

[40] Millar 1964: 190. Jones 1986: 89, notes that Lucian is the first Greek to identify himself with the Romans in this way.

[41] See Millar 1964: 34 and passim.

[42] On Dio's intended readership, see Aalders 1986: 290–291.

impression that his history was in some sense "official," or at least based closely on Latin sources.[43]

Beyond the problem of Dio's style, the lack of a Latin text in Clusters III and IV is just as interesting as the presence of the Greek works in Cluster II. Applied linguists have observed that transference need not be bidirectional.[44] In our study, we failed to find bidirectionality of the kind seen in Dio—though, in the case of Ammianus Marcellinus we have a Greek writing in a Latin style that in some other, unrelated respects does betray Graecisms.[45] We were especially curious to see if in the passages of Latin writers like Aulus Gellius and Cicero differences could be detected between texts with Greek sources and those with Roman sources. We were unable to find any such differences: even the passages with Greek sources are written in a normal Latin style as far as the relative position of direct object and governing verb is concerned.[46]

In an important article about word-order transfer, Odlin concludes with mostly linguistic explanations of the data he considered, which were quite different from the Latin and Greek texts we have been studying. Our limited study suggests that some but not all explanations are linguistic. On the linguistic level, it was surely the greater consistency of Latin SOV word order that helped the Latin pattern to prevail over the more flexible Greek positioning of the verb and direct object.[47] This was true not only for Roman authors writing Latin with a Greek

[43] See Millar 1964: 34.

[44] Cf. Andersen 1983: 191–192; Jordens 1983; and the literature cited by Gass and Selinker 1983: 14.

[45] See Frischer 1996: 113 (on Ammianus' "sentence" length); Rosen 1982.

[46] For example, we found no differences between a very Roman passage such as Cic. *De Leg.* 3.18–38 and a very Greek-inspired passage such as *De Off.* 1.11–30. Similarly, the works of Aulus Gellius with probable Greek and Roman sources did not differ to any detectable degree, thus confirming Steinmetz's observations that Gellius "does not translate [Greek] word for word but clause for clause, and he shapes his clauses according to the requirements of Latin" (Steinmetz 1992: 207, about Gellius' translation of a passage from Plato's *Symposium* [our translation]). Steinmetz finds that Gellius also observes Latin word order in translations of Aeschines (p. 205), of a letter of King Philip of Macedon to Aristotle (p. 208), and of correspondence between Aristotle and Alexander the Great (p. 209). The passages in Aulus studied with probable Roman sources are: 6.2–3; those studied with probable Greek sources are: 1.1–3. For a recent study of the adaptive linguistic tools used by Roman writers to Latinize the Greek texts they were translating see Rosén 1996 with the literature there cited.

[47] Odlin 1990: 109, notes another problem that should be mentioned: the observational difficulty that when a language, like Greek, has a flexible word order, it is difficult to

source before them (like Aulus Gellius or Cicero) but also for a Greek author like Ammianus Marcellinus writing in Latin. It was evidently normally easy for both Greeks and Romans to recognize and to respect the tendency of Latin to place the verb at the end of the clause.[48] However, we would supplement Odlin's purely linguistic explanation with Thomason and Kaufman's thesis that "it is the social context, not the structure of the languages involved, that determines the direction and the degree of interference."[49] Certainly, in the interesting case of the Greek translation of the *Res Gestae* and other official documents, where we encountered the Roman chancellery's habit of translating Latin into Greek through quasi-relexification, we found at work either Roman scrupulosity in legal matters or a sociological factor of linguistic hegemony. Finally, in the case of Cassius Dio we saw the operation of a psycholinguistic or sociolinguistic cause for word-order transference: Dio's conscious or unconscious presentation of himself as a Roman.[50]

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determine whether any given case of SOV is normal or an instance of the influence of Latin.

[48] On such metalinguistic awareness see Odlin 1990: 109–110.

[49] Thomason and Kaufman 1988: 19.

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APPENDIX I: Authors, Texts, Editions, Concordances of Clusters, and Numeric Codes

Author	Text	MCB	OB	MCA	OA	Total	Cluster (1,2)[1]	Cluster (I–IV)[2]	Numeric code[3]
Ammianus Marcellinus	18[4]	39.00	43.00	6.00	12.00	100.00	1	I	L.1
Ammianus Marcellinus	22[5]	41.00	41.00	12.00	6.00	100.00	1	I	L.2
Aulus Gellius	*Misc.* (Greek sources)[6]	35.00	52.00	8.00	5.00	100.00	1	I	L.4
Aulus Gellius	*Misc.* (Latin sources)[7]	37.00	49.00	10.00	4.00	100.00	1	I	L.5
Aulus Hirtius	*Bell. Gall.* 8[8]	37.00	45.00	7.00	11.00	100.00	1	I	L.6
Caesar	*Bell. Gall.* 1[9]	48.00	48.00	3.00	1.00	100.00	1	I	L.7
Caesar	*Bell. Gall.* 3[10]	46.00	51.00	3.00	0.00	100.00	1	I	L.8
Cicero	*Ad Att.*[11]	35.00	46.00	10.00	9.00	100.00	1	I	L.11
Cicero	*Brutus*[12]	24.00	57.00	6.00	13.00	100.00	1	I	L.12
Cicero	*In Cat.*[13]	35.00	40.00	19.00	6.00	100.00	1	I	L.21
Cicero	*De Off.* 1a[14]	27.00	53.00	9.00	11.00	100.00	1	I	L.15
Cicero	*De Off.* 1b[15]	26.00	55.00	6.00	13.00	100.00	1	I	L.16
Cicero	*De Off.* 2a[16]	24.00	53.00	13.00	10.00	100.00	1	I	L.17
Cicero	*De Off.* 2b[17]	30.00	53.00	9.00	8.00	100.00	1	I	L.18
Cicero	*De Off.* 3a[18]	23.00	56.00	11.00	10.00	100.00	1	I	L.19
Cicero	*De Off.* 3b[19]	30.00	55.00	9.00	6.00	100.00	1	I	L.20
Cicero	*De Nat. Deor.*[20]	26.00	60.00	13.00	1.00	100.00	1	I	L.14
Cicero	*De Leg.*[21]	27.00	49.00	16.00	8.00	100.00	1	I	L.13

Author	Text	MCB	OB	MCA	OA	Total	Cluster (1,2)[1]	Cluster (I–IV)[2]	Numeric code[3]
Pliny the Elder	*NH* 34[22]	31.58	41.05	18.95	8.42	100.00	1	I	L.24
Pliny the Elder	*NH* 36[23]	40.82	35.71	12.25	11.22	100.00	1	I	L.25
Pliny (Jun.)	*Epist.*[24]	40.00	34.00	20.00	6.00	100.00	1	I	L.26
Seneca	*De Clem.* 1[25]	38.14	42.27	14.43	5.16	100.00	1	I	L.27
Seneca	*Epist.*[26]	36.00	45.00	11.00	8.00	100.00	1	I	L.28
SHA	*Hadrian*[27]	38.00	45.00	4.00	13.00	100.00	1	I	L.29
SHA	*Marcus Aurelius*[28]	42.00	41.00	12.00	5.00	100.00	1	I	L.30
Tacitus	*Ann.* 1[29]	47.00	42.00	5.00	6.00	100.00	1	I	L.32
Varro	*Res Rust.*[30]	19.00	55.00	7.00	19.00	100.00	1	I	L.34
Augustus	*Res Gestae* (Latin)[31]	63.00	27.00	6.00	4.00	100.00	1	II	L.3
Augustus	*Res Gestae* (Greek)[32]	65.00	21.00	7.00	7.00	100.00	1	II	G.1
Cassius Dio	36[33]	55.00	27.00	9.00	9.00	100.00	1	II	G.2
Cassius Dio	40[34]	60.00	25.00	15.00	0.00	100.00	1	II	G.3
Cassius Dio	54[35]	75.49	14.71	5.88	3.92	100.00	1	II	G.4
Cassius Dio	59[36]	74.23	15.46	8.25	2.06	100.00	1	II	G.5
Cato	*De Agric.*	58.00	38.00	3.00	1.00	100.00	1	II	L.9
Celsus	*De Med.* 2	54.00	33.00	12.00	1.00	100.00	1	II	L.10
Livy	3[37]	49.44	30.34	14.61	5.62	100.00	1	II	L.22
Livy	21[38]	48.45	29.90	16.50	5.16	100.00	1	II	L.23
Tacitus	*Agricola*[39]	65.31	13.27	9.18	12.25	100.00	1	II	L.31
Tacitus	*Hist.* 1[40]	53.00	40.00	6.00	1.00	100.00	1	II	L.33

Author	Text	MCB	OB	MCA	OA	Total	Cluster (1,2)[1]	Cluster (I–IV)[2]	Numeric code[3]
Dionysius of Halicarnassus	*Rom. Ant.* 2[41]	37.63	22.77	14.85	24.75	100.00	2	III	G.6
Herodotus	1[42]	20.00	25.00	25.00	30.00	100.00	2	III	G.7
Herodotus	2[43]	20.00	35.00	21.00	24.00	100.00	2	III	G.8
Marcus Aurelius	3[44]	28.00	40.00	5.00	27.00	100.00	2	III	G.9
Marcus Aurelius	8[45]	28.00	31.00	21.00	20.00	100.00	2	III	G.10
Musonius Rufus	*Discourses*[46]	23.71	23.71	21.65	30.93	100.00	2	III	G.11
Plutarch	*Alcib.*[47]	25.00	24.00	18.00	33.00	100.00	2	III	G.14
Plutarch	*Alex*[48]	15.00	40.00	14.00	31.00	100.00	2	III	G.15
Plutarch	*Cicero*[49]	22.00	37.00	12.00	29.00	100.00	2	III	G.17
Plutarch	*Dinner*[50]	14.00	44.00	12.00	30.00	100.00	2	III	G.12
Plutarch	*Quaest. Rom.*[51]	33.00	25.00	22.00	20.00	100.00	2	III	G.20
Plutarch	*Sertorius*[52]	23.00	43.00	7.00	27.00	100.00	2	III	G.18
Plutarch	*Themist.*[53]	29.00	29.00	21.00	21.00	100.00	2	III	G.19
Polybius	1[54]	21.00	26.00	17.00	36.00	100.00	2	III	G.21
Polybius	2[55]	20.00	24.00	9.00	47.00	100.00	2	III	G.22
Thucydides	2[56]	25.00	35.00	24.00	16.00	100.00	2	III	G.23
Xenophon	*Anab.* 1[57]	17.00	20.00	25.00	38.00	100.00	2	III	G.26
Plutarch	*Camillus*[58]	40.00	21.00	28.00	11.00	100.00	2	IV	G.16
Plutarch	*Quaest. Graec.*[59]	50.00	12.00	30.00	8.00	100.00	2	IV	G.13
Thucydides	5[60]	40.00	13.00	36.00	11.00	100.00	2	IV	G.24
Thucydides	7[61]	37.86	17.48	37.86	6.80	100.00	2	IV	G.25

[1] See Figure A.
[2] See Figure B.
[3] See Figures A, B, and D.
[4] 18.1.1–18.3.6 (ed. J. C. Rolfe).
[5] 22.1.1.–22.7.3 (ed. J. C. Rolfe).
[6] Passages selected because they probably had Greek sources are the following: *Noctes Atticae* 1.1, 1.2, 1.3, 1.5 (ed. J. C. Rolfe).
[7] *Noctes Atticae* 6.2.1–6.3.35 (ed. J. C. Rolfe).
[8] 8.1–8.14 (ed. H. J. Edwards).
[9] 1.1–1.12 (ed. R. L. A. Du Pontet).
[10] 3.1–3.13 (ed. R. L. A. Du Pontet).
[11] Letters 367, 367B, 368, 408, 409 (ed. D. R. Shackleton Bailey).
[12] 75–110 (ed. G. L. Hendrickson).
[13] In Catilinam 1.1.–17 (ed. C. Macdonald).
[14] *De officiis* 1.11–1.30 (ed. W. Miller).
[15] *De officiis* 1.32–1.63 (ed. W. Miller).
[16] *De officiis* 2.11–2.35 (ed. W. Miller).
[17] *De officiis* 2.36–2.55 (ed. W. Miller).
[18] *De officiis* 3.15–35 (ed. W. Miller).
[19] *De officiis* 3.37–3.55 (ed. W. Miller).
[20] *De Natura Deorum* II.lxvii.168–III.i.3; III.i.4–vii.18 (ed. H. Rackham).
[21] *De Legibus* 3.18–38 (ed. C. W. Keyes).
[22] *NH* 34.9.15–34.19.49 (ed. H. Rackham).
[23] *NH* 36.4.15–36.5.44 (ed. H. Rackham).
[24] *Epist.* 3.1.1.–5.4 (ed. B. Radice).
[25] *De Clementia* 1.1.1.–1.5.6 (ed. J. W. Basore).
[26] *Epistulae ad Lucilium* 1.1.1–1.4.2 (ed. R. M. Grummere).
[27] *Vita Hadriani* 1.1.–7.9 (ed. D. Magi).
[28] *Vita Marci Antonini* 1.1–8.2 (ed. D. Magi).
[29] *Annales* 1.1–1.10 (ed. J. Jackson).
[30] *Res Rusticae* 2.1–24 (eds. W. D. Hooper and H. B. Ash).
[31] *Res Gestae* 1.1–4.19 (ed. E. Weber).
[32] *Res Gestae* 1.1–10.20 (ed. E. Weber).
[33] 36.1.1–36.12.1 (ed. E. Cary).
[34] 40.16.1–40.29.2 (ed. E. Cary).
[35] 54.7.1–54.11.2 (ed. E. Cary).
[36] 59.2.1–59.5.3 (ed. E. Cary).
[37] 3.1.1–3.8.5 (ed. B. O. Foster).
[38] 21.1.1–21.9.4 (ed. B. O. Foster).
[39] *Agricola* 1.1–10.6 (ed. M. Hutton and R. M. Ogilvie).
[40] *Historiae* 1.1–1.15 (ed. C. H. Moore).
[41] *Rom. Ant.* 2.1.1.–2.6.1 (ed. E. Cary).
[42] *Hist.* 1.1.–1.13 (ed. A. D. Godley).
[43] *Hist.* 2.1–2.14 (ed. A. D. Godley).
[44] *Medit.* 3.1–3.10 (ed. A. S. L. Farquharson).
[45] *Medit.* 8.1–8.41 (ed. A. S. L. Farquharson).
[46] 1–3 (ed. C. E. Lutz).

[47] 26.1–30.3 (ed. B. Perrin).
[48] 1.1–6.2 (ed. B. Perrin).
[49] 7.1–12.2 (ed. J. L. Moles).
[50] *Convivium septem sapientium*, *Moralia* 158.15–162.19 (ed. F. C. Babbitt).
[51] *Moralia* 263E–267A (ed. F. C. Babbitt).
[52] 11.2–16.5 (ed. B. Perrin).
[53] 1.1–7.2 (ed. B. Perrin).
[54] 1.33.1–1.38.8 (ed. W. R. Paton).
[55] 2.25.1–2.30.4 (ed. W. R. Paton).
[56] 2.12–2.21 (ed. C. F. Smith).
[57] 1.1–2.12 (ed. C. L. Brownson).
[58] 1.1–7.2 (ed. B. Perrin).
[59] 291E–294E (ed. F. C. Babbitt).
[60] 5.1.1–5.10.3 (ed. C. F. Smith).
[61] 7.1.1–7.11.4 (ed. C. F. Smith).